

8	A	Work done on sledge lifting up 1.2 m $W_{\text{on sledge}} = mg\Delta h$ $= 15 \times 9.81 \times 1.2$ $= 177 \text{ J}$ No work done on sledge pulling it across ice, because no frictional force on sledge by ice. Work done on sledge pulling it 10 m across snow $W_{\text{on sledge}} = F_{\text{friction}} s \text{ (pulling force is equal in magnitude to } F_{\text{friction}} \text{ at constant speed)}$ $= 60 \times 10$ $= 600 \text{ J}$ No work done on sledge lifting it vertically and moving back at a constant speed to the starting point, because the lifting force is perpendicular to the direction of travel.
9	B	Instantaneous power required to drive the accelerator